

Physics .

MCQs :-

1. B
2. A
3. B
4. D
5. D
6. C
7. A
8. B
9. B
10. C
11. C
12. B

Q: 2(i)

A: 2(i)

(i) SI - base quantities :-

- (1) Length - meter - m
- (2) Mass - Kilogram - kg
- (3) Time - Seconds - s
- (4) Temperature - Kelvin - K
- (5) Electric current - Ampere - A
- (6) Amount of substance - mole - mol
- (7) Light intensity - Candela - Cd

- (i) (a) Area :- m^2 , (b) Velocity :- m/s
(c) force :- $kg \cdot m/s^2$

Q.2(ii)

A:2(ii)

(a) Speed

Velocity

① Measure of the distance covered with the passage of time.

① Measure of the displacement covered with the passage of time.

② It is a scalar quantity.

② It is a vector quantity.

③ It's ~~denoted~~ symbol is s or v .

③ It's symbol is $\vec{s}$ or $\vec{v}$.

④ It's formula is:-
$$v = \frac{s}{t}$$

It's formula is
$$\vec{v} = \frac{\vec{s}}{t}$$

(b) No, the velocity is not constant because the ~~motion~~ direction of motion changes continuously while the car moves in circular path, even though its speed remains constant.

Q: 2(iii)

A: 2(iii)

initial velocity :- 8m/s

final velocity :- 32m/s

time :- 6s

a) acceleration :-
$$\frac{v_f - v_i}{t}$$

$$= \frac{32 - 8}{6}$$

$$= \frac{24}{6} \Rightarrow 4 \text{ m/s}^2$$

b) Distance covered :-

$$\frac{v}{t} = \frac{s}{t} \quad s = vt$$

$$s = 24 \times 6$$
$$= 144 \text{ m.}$$

$$s = v_i \times t + \frac{1}{2} at^2$$

$$= 8 \times 6 + \frac{1}{2} (4) (6)^2$$

$$48 + \frac{4}{2} (6)^2$$

$$48 + \frac{4}{2} (36)$$

$$48 + \frac{36}{7.2} \quad 48 + 7.2$$

$$48 + 36 \quad 120 \text{ m}$$

Q. 2(iv) (or)

A. 2(iv) (or)

Second Law of Newton's:-

The acceleration of an object is directly proportional to the magnitude of net force, in a same direction, is inversely proportional to the mass of an object.

$$F = ma$$

$$m = \frac{F}{a} \quad m = \frac{F}{a}$$

$$m = 45 \text{ kg}$$

8,

$$m = 9 \text{ kg}$$

Q: 2(v)(or)

A: 2(v)(or)

net force :-

The net (resultant) force is overall acting on an object after ~~and~~ adding all forces together, taking their directions into account.

Examples

* $10\text{N} \rightarrow \boxed{+} \leftarrow 6\text{N}$
* Net force :- 4N

* $9\text{N} \rightarrow \boxed{+} \leftarrow 5\text{N}$
Net force :: 4N .

* $15\text{N} \rightarrow \boxed{+} \leftarrow 15\text{N}$
Net force :- 0N
The object is balanced.

Q: 2(vi)(or)

A: 2(vi)(or)

Terminal velocity is the maximum constant velocity speed of a falling object when air resistance becomes equal to its weight.

A skydiver's weight remains constant while air resistance increases as she falls faster. When air resistance equals her weight, the resultant force becomes zero, and she falls at a constant speed called terminal velocity.

Q: 2(vii)

A: 2(vii)

Advantages:-

- ① Friction holds the screw and nails in woods.
- ② Friction b/w tyre and road helps to drive cars.

Disadvantages:-

- ① It slows down moving objects and causes heating of moving parts in machinery.
- ② Produce wear and tear.

Reducing of friction:-

By polishing: if we polish the rough surfaces, they become smooth and friction is reduced.

$$\text{Q: 2(viii)(or)} \\ \text{A: 2(viii)(or)} \\ \text{a) Pressure: } pgh$$

$$P = 2\text{m} \times 10 \times 700 \\ 14,000 \text{ Pa}$$

b) Total pressure :-

$$P = 3\text{m} \times 10 \times 1000 \\ 30,000$$

$$P_1 + P_2 = 14,000 + 30,000 \\ 44,000 \text{ Pa}$$

Q: 2(in) \\ A: 2(in) \\ @ Pressure is defined as force per unit-area. Pressure is represented by 'P'. It's equation is :-

$$P = \frac{F}{A}$$

$$\text{b) } P = \frac{F}{A} \Rightarrow \frac{4500}{0.25} \\ \approx 18,000 \text{ Pa}$$

Q.2(n)

Workdone:-

work is said to be done when force displaces in it's own direction.

formula:- $F \cdot s$

$$W = FS(\cos)(\theta)$$

$$W = f6(0.77)$$

$$W = 4.62 F$$

Q.2(xii)(or)

A.2(xii)(or)

Thermal Physics:-

is the collective study of statistical mechanics, thermodynamics and kinetic theory of gases.

Nuclear Physics:-

is the study of the structure, properties and behaviour of atomic nuclei.

Q.3(i)

A:3(i)

a) Derivation of $v_f = v_i + at$

from the actual definition of acceleration :-

$$a = \frac{v_f - v_i}{t}$$

$$at = v_f - v_i$$

$$v_f = v_i + at$$

b) Given :- $v_i = 7 \text{ m/s}$
 $v_f = 27 \text{ m/s}$
 $t = ?$
 $s = 136 \text{ m}$

i) Acceleration :-

$$\begin{aligned} (v_f)^2 &= (v_i)^2 + 2(a)(s) \\ (27 \text{ m/s})^2 &= (7)^2 + 2(a)(136) \\ 729 &= 49 + 272a \\ 729 - 49 &= 272a \\ 680 &= 272a \\ 680 \div 272 &= a \\ 2.5 \text{ m/s}^2 &= a \end{aligned}$$

(ii) find time.

$$v_f = v_i + at$$

$$27 = 7 + 2.5t$$

$$27 - 7 = 2.5t$$

$$20 = 2.5t$$

$$\frac{20}{2.5} = t$$

$$8s$$

$$8s = t$$

Q: 3 (ii) (or)

A: 3 (ii) (or)

(a) Law of conservation of momentum
The total momentum of an isolated system remains constant if no external force acts on it.

(b) Using conservation of momentum:

$$m_1 v_{1i} + m_2 v_{2i} = (m_1 + m_2) v_f$$

$$1000(18) + 1500(-6) = (2500) v_f$$

$$18000 - 9000 = 2500 v_f$$

$$9000 = 2500 v_f$$

$$v_f = 3.6 \text{ m/s}$$

② Loss in kinetic energy.

$$KE_i = \frac{1}{2}(1000)(18)^2 + \frac{1}{2}(1500)(6)^2$$
$$= 162000 + 27000 = 189000 \text{ J}$$

$$KE_f = \frac{1}{2}(2500)(3.6)^2$$

$$\frac{1250 \times 12.96}{16200 \text{ J}}$$

loss in KE :-

$$189000 - 16200 = 172800 \text{ J}$$

Q: 3(iii) (or)

A: 3(iii) (or)

Hooke's law :-

Within the elastic limit, the extension of a spring is directly proportional to the applied force.

$$F = kx$$

spring constant (k) :-

The force required to produce unit extension in a

Spring
SI unit :- N/m

① force $\therefore F = 10\text{N}$

Extension: $x = 5\text{cm} = 0.05\text{m}$

maximum extension :- $12\text{cm} = 0.12\text{m}$

② Spring constant :-

$$k = \frac{F}{x}$$

$$= \frac{10}{0.05} = 200\text{N/m}$$

③ Maximum load :-

$$F = kx$$

$$= 200 \times 0.12$$

$$24\text{N}$$

$$Q: 2(iv)$$

$$A: 3(iv)$$

(a) Statement :-

Energy can neither be created nor be destroyed; it can only be transformed from one form to another.

(b)(i) $PE = mgh$

$$PE = 6 \times 10 \times 45$$

$$2700 \text{ J}$$

(ii) $KE = PE = 2700 \text{ J}$

Speed :- $KE = \frac{1}{2} mv^2$

$$2700 = \frac{1}{2} \times 6 (v)^2$$

~~$$2700 \times 2 = 6v^2$$~~

$$\frac{2700}{3} = v^2$$

$$\sqrt{900} = v$$

$$30 = v \text{ m/s}$$

① Actual speed = 24 m/s

$$KE = \frac{1}{2}mv^2$$

$$= \frac{1}{2} (3)(24)^2$$

$$= 3(24)^2$$

1728 J

work done against air resistance

$$2700 - 1728$$

$$972$$